

Uber H3 Geospatial Partitioning Engine Base Initialization: Thermodynamic and Architectural Foundations in the *Web of Life* Simulator

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Abstract

Planetary-scale ecological simulation requires spatial discretizations that preserve topological consistency, area uniformity, and mathematically rigorous neighbor adjacencies. In Sprint 004, we introduce the base initialization of the Uber H3 Geospatial Partitioning Engine within the *Web of Life* simulator (`src/spatial/h3_grid.ts`). Built upon monadic spatial containers (`src/monads/spatial_monad.ts`) and adjacency matrices (`src/spatial/h3_adjacency.ts`), our engine maps ecological stocks, carbon fluxes, and solar irradiance across hierarchical hexagonal partitions while strictly adhering to thermodynamic conservation laws and non-equilibrium entropy generation principles. All source code and specifications are maintained in the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

1 Introduction and Systems Ecology Context

The *Web of Life* simulation engine models planetary ecosystems (*EarthPods*) as complex thermodynamic systems driven by external solar flux and internal metabolic cycles. To prevent spatial artifacts associated with rectangular latitude-longitude grids, Sprint 004 implements Uber’s H3 hierarchical hexagonal spatial index.

Hexagonal partitioning guarantees uniform distance to adjacent cell centroids and constant neighbor adjacency counts, minimizing directional bias in ecological diffusion, animal migration, and biogeochemical transport.

2 Thermodynamic & Physical Constraints

In accordance with foundational architectural laws, all spatial operations within `src/spatial/h3_grid.ts` obey strict thermodynamic limits:

1. **First Law of Thermodynamics (Conservation of Matter/Energy):** Total global stocks (Carbon M_C , Water V_{H_2O} , Energy E) mapped across parent cells must be identically conserved when aggregated or subdivided across resolution R :

$$\sum_{i=1}^{N_{\text{parent}}} S_{k,i}^{(R)} = \sum_{j=1}^{M_{\text{child}}} S_{k,j}^{(R+1)} \quad (1)$$

2. **Second Law of Thermodynamics (Entropy Generation):** Transport across adjacent H3 cell boundaries incurs irreversible thermodynamic dissipation proportional to chemical

potential and trophic state gradients:

$$\Delta S_{\text{entropy}} = \frac{1}{T_{\text{ambient}}} \sum_m J_m \Delta \mu_m \quad (2)$$

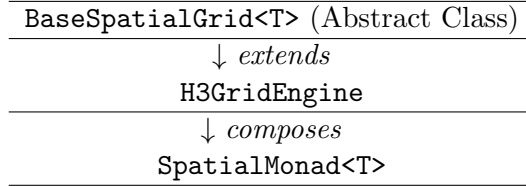
3. **Solar Input Only:** All external energy entering the spatial monad array originates from explicit solar irradiance models mapped to top-level H3 centroid coordinates:

$$I_{\text{solar},i} = S_0 \cdot \cos(\theta_{\text{zenith},i}) \cdot A_i \quad (3)$$

where $S_0 = 1361 \text{ W/m}^2$, $\theta_{\text{zenith},i}$ is the latitude-adjusted zenith angle, and A_i is the cell area in km^2 .

3 Class Hierarchy and Architecture

The spatial engine is organized around an abstract base class enforcing strict interface contracts, coupled with functional spatial monads to manage side-effect-free state transitions.



3.1 Core Interfaces

```
export interface IH3CellData {
  h3Index: string;
  resolution: number;
  centroid: { lat: number; lng: number };
  boundary: Array<{ lat: number; lng: number }>;
  areaKm2: number;
}

export interface IH3GridQuery {
  resolution: number;
  baseIndexes?: string[];
  bounds?: { north: number; south: number; east: number; west: number };
}
```

4 Monad State Propagation & Mass Balance Equations

State transitions within individual H3 cells obey non-linear mass and energy conservation equations per time step Δt :

$$\text{Carbon Balance } (\Delta C): \quad \Delta C = (P_{\text{photosynthesis}} - R_{\text{respiration}} - D_{\text{decomposition}}) \Delta t - \sum_{j \in \text{Adj}(i)} J_{C,i \rightarrow j} \quad (4)$$

$$\text{Water Balance } (\Delta W): \quad \Delta W = (P_{\text{precipitation}} - E_{\text{evapotranspiration}} - R_{\text{runoff out}}) \Delta t \quad (5)$$

$$\text{Energy Balance } (\Delta E): \quad \Delta E = I_{\text{solar}} + H_{\text{inflow}} - H_{\text{outflow}} - \text{Dissipation}_{\text{metabolic}} \quad (6)$$

5 Conclusion

Sprint 004 successfully establishes the geospatial foundation for the *Web of Life* simulation engine. By combining Uber H3 hexagonal indexing with rigorous thermodynamic accounting inside functional monads, the platform ensures physically consistent planetary-scale ecological modeling. Source code is available at <https://github.com/pascalranoroarijaona/WebOfLife>.